Local water and new development

How could a new building affect local water supplies and rivers?

Our current tool estimates water used to cool equipment.

For town planners, researchers, and water providers.

Current test data: generated examples.
Next: test measured water use at actual sites.

How the program works

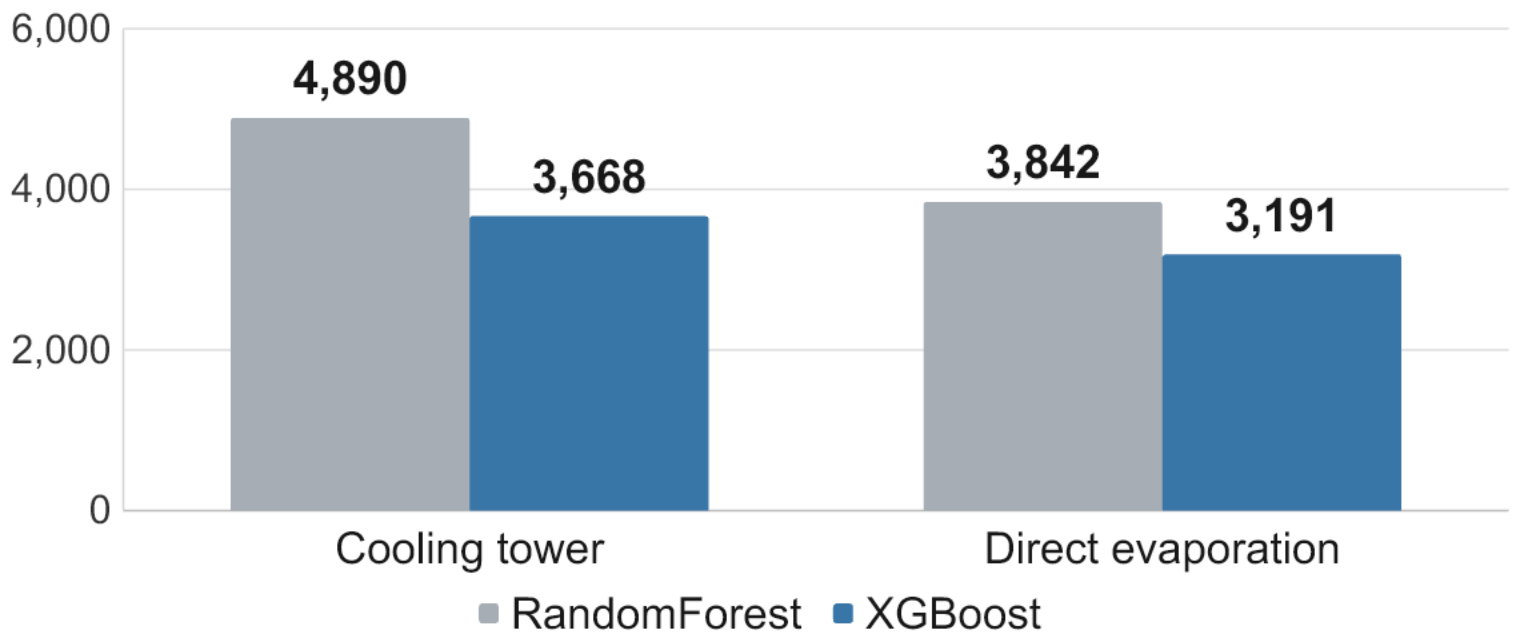
1. Inputs	2. Estimate	3. Check	4. Result
Equipment + place Weather + river	Learn from examples Predict daily water use	Cooling limits Local permit quotes	Gallons/day + range Warnings + sources

The model learns patterns. Permit search compares words and checks place.

The program returns quotes. A person decides how a permit applies.

Test 1: Which model makes smaller errors?

Average error in gallons per day. Lower is better.



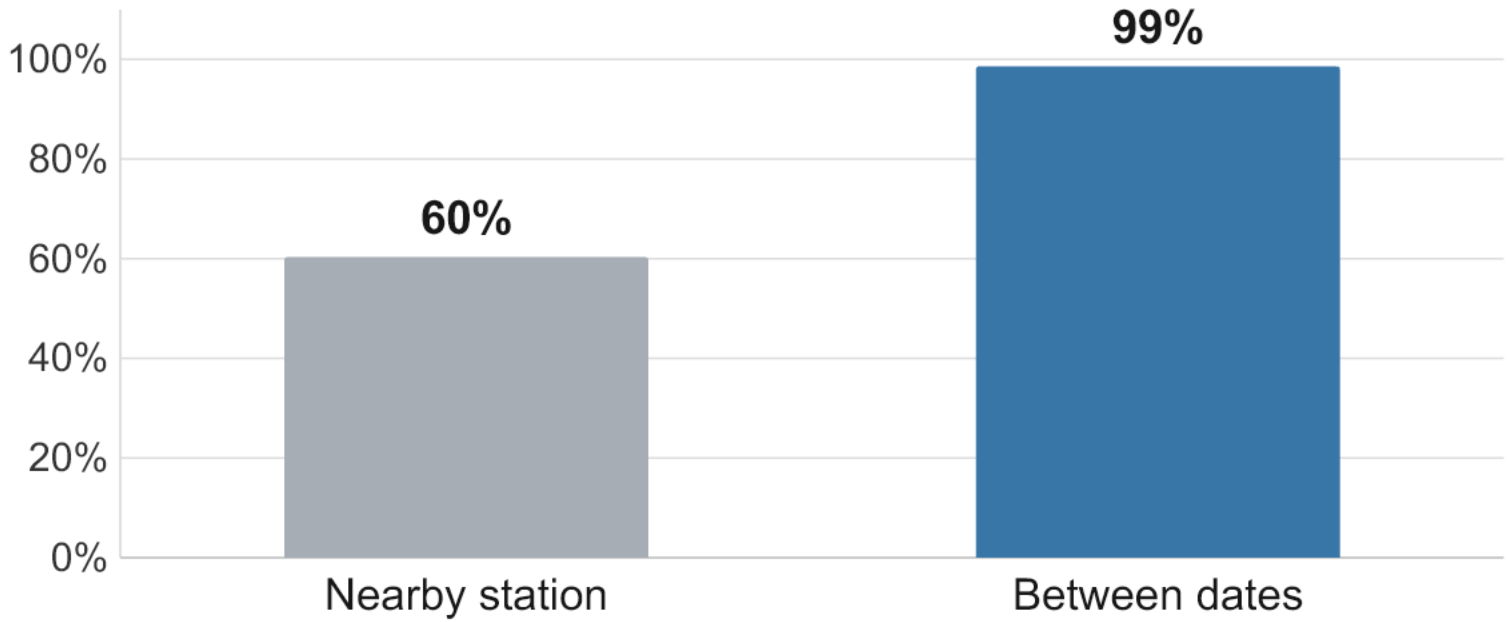
Both models learn from examples.

XGBoost made smaller errors in this test.

E1 / regression-comparison.csv / 333 test examples per cooling method

Test 2: What if river readings are missing?

Bad nearby readings: share of changes within the allowed limit



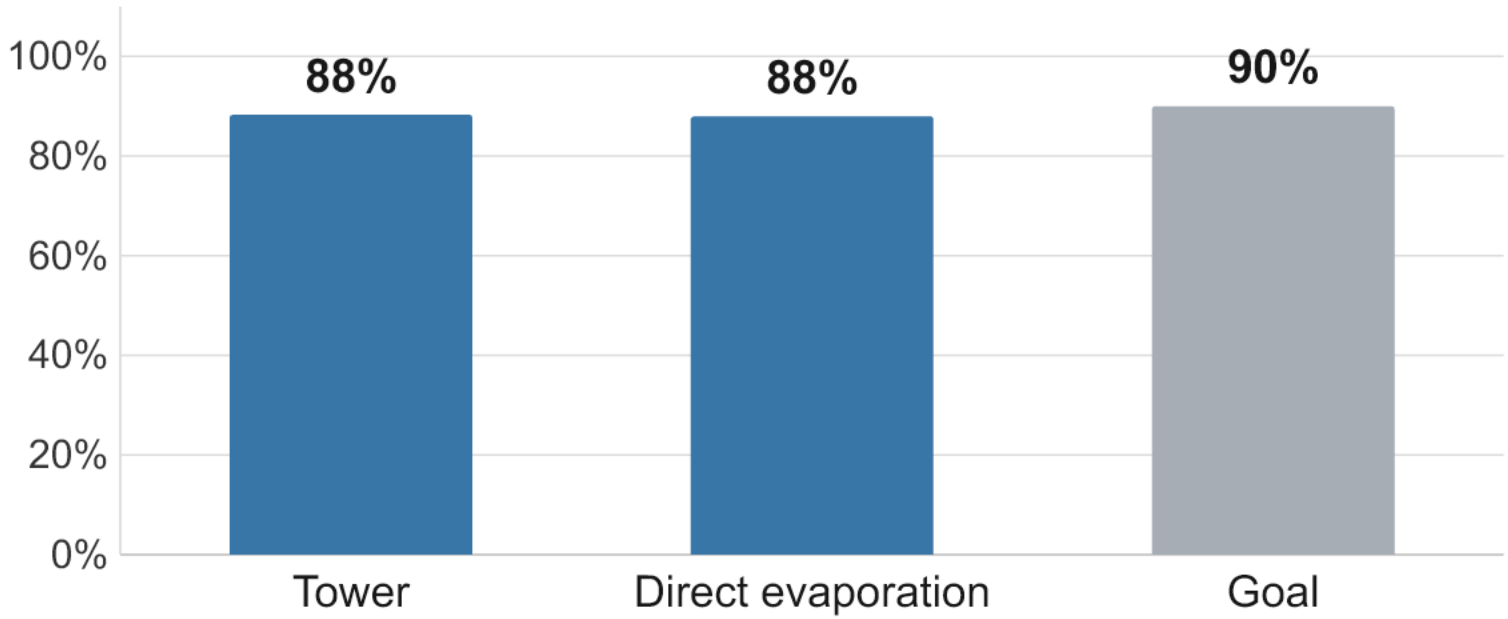
Within 50,000 gallons/day of the no-gap result.

Nearby readings need to represent the same conditions.

With similar stations, both methods stayed close in the main test.

Test 3: Did the prediction range work?

Share of test answers inside the low-to-high range



Goal: 90 out of 100.
Got: about 88.

Use separate data
to choose the model
and set its range.

E5 / interval-coverage.csv / Tower: 294 of 333 / Direct evaporation: 293 of 333

What we will keep and improve

Keep (Adopt)	Model comparison, physical checks, local quotes.
Improve (Modify)	Missing readings and prediction ranges.
Reject	A good model fit as a promise of accuracy.
Wait (Defer)	More complicated AI until we have measured examples.

Next: train on some sites and test on different sites.